

JONATHAN EARP

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linkedin.com/in/earpjonathan | github.com/earpjonathan | FAA Part 107 Remote Pilot | U.S. Citizen

I mostly write Python, with some Rust and JavaScript. At LBNL I wrote the three-stage labeling pipeline behind 40,000+ image pairs. On my own I've built a pipeline from drone video to a 3D Gaussian Splatting model, and one that renders TV clips as Desmos animations (4.4M plays).

EDUCATION

University of Edinburgh

Edinburgh, UK · 2026 – 2031

MEng Electrical and Mechanical Engineering, Year 1

The Head-Royce School

Oakland, CA · Jun 2026

GPA 3.80 unweighted, 4.38 weighted. Honors Calculus, Advanced Physics, Advanced Chemistry, Data Science

PROJECTS

3D Reconstruction from FPV Flight Data — S.P.L.O.O.G.E | Python, COLMAP, Rust, WebGL

Aug 2026

- Built a pipeline from raw DJI O4 Pro video and gyro telemetry to one 3D Gaussian Splatting model of two flights six days apart: 7,470 of 7,471 frames registered at 0.82 px mean reprojection error.
- Estimated the uncalibrated lens focal length by regressing optical flow against gyro rotation ($R^2 = 0.967$), and cut video-to-render misalignment from 72.3 px to 0.89 px by undistorting the footage.
- Confirmed 13,482 verified image matches linked the two sites before committing to an 18-hour mapping run.
- Traced a rendering defect to the viewer's near-plane cull after measuring four other fixes, including a depth-distortion loss I implemented in Rust. Holes fell from 51.2% to 4.4%, with no regressions across 56 test poses.

Automated Video Rendering Pipeline — P.E.T.E.R | Python, OpenCV, Playwright, ffmpeg

2026

- Wrote a 4,100-line Python pipeline that renders video clips as Desmos equations and uploads them. 4.4M plays.
- Profiled rendering (94% capture, 4% tracing) and parallelized capture for a 1.55x speedup, then found a timing bug corrupting 39 of 40 frames and fixed it by sweeping the wait threshold until output converged.

FPV and UAS Builds — Personal | Betaflight, ArduPilot, ExpressLRS

2023 – Present

Head-Royce VEX Robotics — Team Captain

Oakland, CA · 2022 – 2026

EXPERIENCE

Lawrence Berkeley National Laboratory — Student Research Assistant

Berkeley, CA · Apr 2024 – Present

StoveCam: low-cost camera sensor for an indoor air quality and childhood asthma study

- Wrote a three-stage Python labeling pipeline: an on-device brightness gate, a low-cost cooking screen, then Gemini vision for food, method, burner and cookware. 40,000+ RGB/thermal pairs from three homes at \$0.018 per analyzed pair.
- Validated the output against 506 hand-graded pairs: mean score 8.1/10, with zero burner or cooking-method errors. Most failures traced to residual-heat false positives. Abstract submitted to ISES 2026.

University of Pennsylvania, Singh Center for Nanotechnology — ESAP Participant

Philadelphia, PA · Summer 2025

SKILLS

Programming: Python (NumPy, OpenCV, Playwright), Rust, COLMAP, JavaScript, WebGL, Arduino, Raspberry Pi / Linux, Git

Controls & sensors: PID tuning, IMU and gyro data, quaternion orientation, sensor calibration, flight-log analysis

Electrical & avionics: Brushless motors, STM32 ESCs, soldering, Betaflight, ArduPilot, ExpressLRS, analog and DJI video

CAD & fabrication: Onshape, Fusion 360, 3D printing, CNC, laser cutting, bandsaw, drill press, photolithography

LEADERSHIP

- Medical Advocacy, President.** Ran a blood drive that brought in 120 donors, and set up a partnership with a local hospital.
- Eagle Scout, Senior Patrol Leader.** Taught a month-long robotics class to elementary and middle school students for my Eagle project, and left the curriculum behind so someone else could run it again.